(FP1) 1. Roots of polynomial equations

NO. 1

The roots of the cubic equation

$$x^3 - 5x^2 + 13x - 4 = 0$$

are α, β, γ.

i. Find the value of $\alpha^2 + \beta^2 + \gamma^2$. [3]

.....

ii. Find the value of $\alpha^3 + \beta^3 + \gamma^3$. [2]

.....

NO. 2

The roots of the equation

$$x^3 + px^2 + qx + r = 0$$

are α, 2α, 4α, where p, q, r and α are non-zero real constants.

i. Show that [4]

$$2p\alpha + q = 0.$$

.....

ii. Show that [2]

$$p^3 r - q^3 = 0.$$

.....

NO. 3

It is given that the equation

$$x^3 - 21x^2 + kx - 216 = 0,$$

where k is a constant, has real roots a, ar and ar^{-1}.

i. Find the numerical values of the roots. [6]

.....

ii. Deduce the value of k. [2]

.....

NO. 4

The equation

$$x^3 - x + 1 = 0$$

has roots α, β, γ.

i. Use the relation $x = y^{\frac{1}{3}}$ to show that the equation

$$y^3 + 3y^2 + 2y + 1 = 0$$

has roots $\alpha^3, \beta^3, \gamma^3$. Hence write down the value of $\alpha^3 + \beta^3 + \gamma^3$. [3]

.....

Let $S_n = \alpha^n + \beta^n + \gamma^n$

ii. Find the value of S_{-3}. [2]

.....

iii. Show that $S_6 = 5$ and find the value of S_9. [4]

.....

NO. 5

The equation $x^3 + 2x^2 + x + 7 = 0$ has roots α, β, γ.

i. Use the relation $x^2 = -7y$ to show that the equation

$$49y^3 + 14y^2 - 27y + 7 = 0$$

has roots $\frac{\alpha}{\beta\gamma}, \frac{\beta}{\gamma\alpha}, \frac{\gamma}{\alpha\beta}$. [4]

.....

ii. Show that $\frac{\alpha^2}{\beta^2\gamma^2} + \frac{\beta^2}{\gamma^2\alpha^2} + \frac{\gamma^2}{\alpha^2\beta^2} = \frac{58}{49}$. [3]

.....

iii. Find the exact value of $\frac{\alpha^3}{\beta^3\gamma^3} + \frac{\beta^3}{\gamma^3\alpha^3} + \frac{\gamma^3}{\alpha^3\beta^3}$. [2]

.....

NO. 6

The roots of the cubic equation $x^3 + 2x^2 - 3 = 0$ are α, β and γ.

i. By using the substitution $y = \frac{1}{x^2}$, find the cubic equation with roots $\frac{1}{\alpha^2}$, $\frac{1}{\beta^2}$ and $\frac{1}{\gamma^2}$. [3]

.....

ii. Hence find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$. [1]

.....

iii. Find also the value of $\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\gamma^2\alpha^2}$. [1]

.....

NO. 7

The cubic equation $2x^3 + 5x^2 - 6 = 0$ has roots α, β, γ.

a. Find a cubic equation whose roots are $\frac{1}{\alpha^3}, \frac{1}{\beta^3}, \frac{1}{\gamma^3}$. [3]

b. Find the value of $\frac{1}{\alpha^6} + \frac{1}{\beta^6} + \frac{1}{\gamma^6}$. [3]

c. Find also the value of $\frac{1}{\alpha^9} + \frac{1}{\beta^9} + \frac{1}{\gamma^9}$. [2]

NO. 8

The cubic equation $x^3 + x^2 + 3x + 3 = 0$ has roots α, β, γ.

a. Find the value of $\alpha^2 + \beta^2 + \gamma^2$. [2]

b. Show that $\alpha^3 + \beta^3 + \gamma^3 = 1$. [2]

c. Use standard results from the list of formulae (MF19) to show that

$$\sum_{r=1}^{n}((\alpha+r)^3 + (\beta+r)^3 + (\gamma+r)^3) = n+\frac{1}{4}n(n+1)(an^2+bn+c),$$

where a, b and c are constants to be determined. [6]

NO. 9

The cubic equation $2x^3 - 4x^2 + 3 = 0$ has roots α, β, γ. Let $S_n = \alpha^n + \beta^n + \gamma^n$.

a. State the value of S_1 and find the value of S_2. [3]

b. i. Express S_{n+3} in terms of S_{n+2} and S_n. [1]

ii. Hence, or otherwise, find the value of S_4. [2]

c. Use the substitution $y = S_1 - x$, where S_1 is the numerical value found in part **(a)**, to find and simplify an equation whose roots are $\alpha+\beta$, $\beta+\gamma$, $\gamma+\alpha$. [3]

d. Find the value of $\frac{1}{\alpha+\beta} + \frac{1}{\beta+\gamma} + \frac{1}{\gamma+\alpha}$.

NO. 10

The equation $x^4 - 2x^3 - 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

a. Find a quartic equation whose roots are α^3, β^3, γ^3, δ^3 and state the value of $\alpha^3 + \beta^3 + \gamma^3 + \delta^3$. [4]

b. Find the value of $\frac{1}{\alpha^3} + \frac{1}{\beta^3} + \frac{1}{\gamma^3} + \frac{1}{\delta^3}$. [3]

c. Find the value of $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$. [2]